IT4010 Research Project – 2025 July

Topic Assessment Form

Project ID : 25-26J-440

1. Topic (12 words max)

A Smart Livestock Monitoring and Dairy Management System to Enhance Productivity in Medium-Scale Farms in Sri Lanka.

2. Research group the project belongs to

SST - Software Systems & Technologies

3. Specialization of the project belongs to

Software Engineering (SE)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Middle-scale cattle farmers in Sri Lanka face significant challenges in effectively managing farm operations due to low technology adoption, limited access to localized knowledge, and manual processes. Key tasks such as cattle health monitoring, Individual Monitoring, disease detection, and environmental control are still performed manually or not at all, resulting in inefficiencies, increased animal stress, missed health risks, and lost productivity. Additionally, language barriers further restrict access to reliable farming knowledge, leading to poor decision-making and suboptimal farm performance [1][2][3].

Lack of RealTime Individual Monitoring and Data Driven Decision Making in Dairy Farming Production - Many dairy farmers struggle because they don't have a way to track each cow's milk production in real-time. Instead, they use manual methods or look at the whole herd's average, which means they miss early signs of problems like late milk peaks, sudden drops, or long dry periods. Without this information, farmers can't make quick and smart decisions about feeding, breeding, or health care. This leads to wasting feed, poor breeding results, and lower milk production. That's why it's important to have a smart system that tracks each cow's performance and helps farmers make better decisions to improve their productivity and profitablity. [4][5].

Farmers lack access to localized, real-time information about cattle care and farming best practices.-Additionally, there is a noticeable gap in the availability of cattle farming knowledge in local languages. This language barrier prevents many farmers from accessing essential

guidelines, best practices, and training resources related to effective cattle farm management. As a result, farmers struggle to handle the various operational aspects of the farm efficiently.[6]

Delayed detection of diseases and limited access to professional veterinary support leads to health risks. - Health management of cattle is another critical issue, particularly in the early detection and identification of common diseases such as Foot and Mouth Disease (FMD). Most farmers are not equipped with the technical knowledge or tools to detect these conditions in their early stages, leading to delayed treatment and potential loss of livestock [7][8][9].

Manual Water Control Systems fail to respond to real-time heat stress conditions in cattle, leading to inefficient cooling, excessive water usage, and a decline in animal health and productivity - Heat stress in dairy cattle leads to reduced milk yield, increased illness, low feed intake, and fertility problems. Existing research use either manual timers or only ambient temperature sensors and activate sprinklers for the whole barn without identifying whether each cow is under stress. These methods are inefficient, waste water, and don't improve individual cattle welfare. [10][11].

Together, these issues highlight the need for an integrated and user-friendly solution that supports middle-scale cattle farmers by automating data collection, offering localized language support, enabling early disease detection, and managing environmental factors efficiently.

[1] F. Hu, M. Zhou, Y. Qin, Q. Zhao, Y. Xin and G. Li, "Novel Cattle Farm Management System Based on IoT and RFID," *2022 3rd Int. Conf. on Computer Science and Management Technology (ICCSMT)*, Shanghai, China, 2022, pp. 60–63, doi: 10.1109/ICCSMT58129.2022.00020.

[2] N. Markov et al., "Smart Dairy Farm – Digitalization and Innovation," Research Institute of Mountain Stockbreeding and Agriculture, Bulgaria, and Institute of Information and Communication Technologies, Bulgarian Academy of Science, Sofia, 2022. [Online].

[3] E. Paunova-Hubenova and E. Trichkova-Kashamova, "Smart Solutions for Control and Management in Livestock Farms," 2021 Big Data, Knowledge and Control Systems Engineering (BdKCSE), Sofia, Bulgaria, 2021, pp. 1–7, doi: 10.1109/BdKCSE53180.2021.9627269.

[4]S. Stankovski, G. Ostojic, I. Senk, M. Rakic-Skokovic, S. Trivunovic, and D. Kucevic, "Dairy cow monitoring by RFID," *Scientia Agricola*, vol. 69, no. 1, pp. 75–80, Feb. 2012, doi: https://doi.org/10.1590/s0103-90162012000100011.

[5]M. Lee, S. Lee, J. Park, and S. Seo, "Clustering and Characterization of the Lactation Curves of Dairy Cows Using K-Medoids Clustering Algorithm," *Animals*, vol. 10, no. 8, p. 1348, Aug. 2020, doi: https://doi.org/10.3390/ani10081348.

[6] K. Herrera, J. Miranda and D. Mauricio, "Milchbot: App to Support the Process of Feeding and Caring for Dairy Cows in Peru," AGRIS on-line Papers in Economics and Informatics, vol. 14, no. 4, pp. 3–10, Dec. 2022.

[7] R. N. Shebiah and S. Arivazhagan, "Deep Learning Based Image Analysis for Classification of Foot and Mouth Disease in Cattle," 2023 5th Int. Conf. on Inventive Research in Computing Applications (ICIRCA), Coimbatore, India, 2023, pp. 701–705, doi: 10.1109/ICIRCA57980.2023.10220765.

[8] R. Maganti, S. Yalamanchili and B. Mediboyina, "Cattle Skin Diseases Classification Using Deep Learning Models," 2024 1st Int. Conf. on Data, Computation and Communication (ICDCC), Sehore, India, 2024, pp. 698–703, doi: 10.1109/ICDCC62744.2024.10961322.

[9] T. Velugoti, S. B. Dasari and S. S. Sultana, "Lumpy Skin Disease Detection Using Deep Learning," 2024 IEEE Students Conf. on Engineering and Systems (SCES), Prayagraj, India, 2024, pp. 1–6, doi: 10.1109/SCES61914.2024.10652331.

[10] E. Liu, L. Liu, Z. Zhang, M. Qu and F. Xue, "An Automated Sprinkler Cooling System Effectively Alleviates Heat Stress in Dairy Cows," *Animals*, vol. 14, 2024, Art. no. 2586, doi: 10.3390/ani14172586.

[11] S. Goswami, "Implementation of Automatic Cooling System for Cattle Shed and Its Effect on Milk Production," *Indian Research Journal of Extension Education*, vol. 21, no. 2&3, pp. 137–143, Apr.–Jul. 2021.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

To address the challenges faced by medium-scale dairy farmers in Sri Lanka, this research proposes a **Smart Dairy Farm Management Platform** that integrates cattle monitoring, health tracking, farmer education, and automated climate control into a unified, mobile-accessible system. Designed specifically for the local context, the platform emphasizes **accessibility, real-time insights, and language support in Sinhala and English**.

The system is built around four core modules:

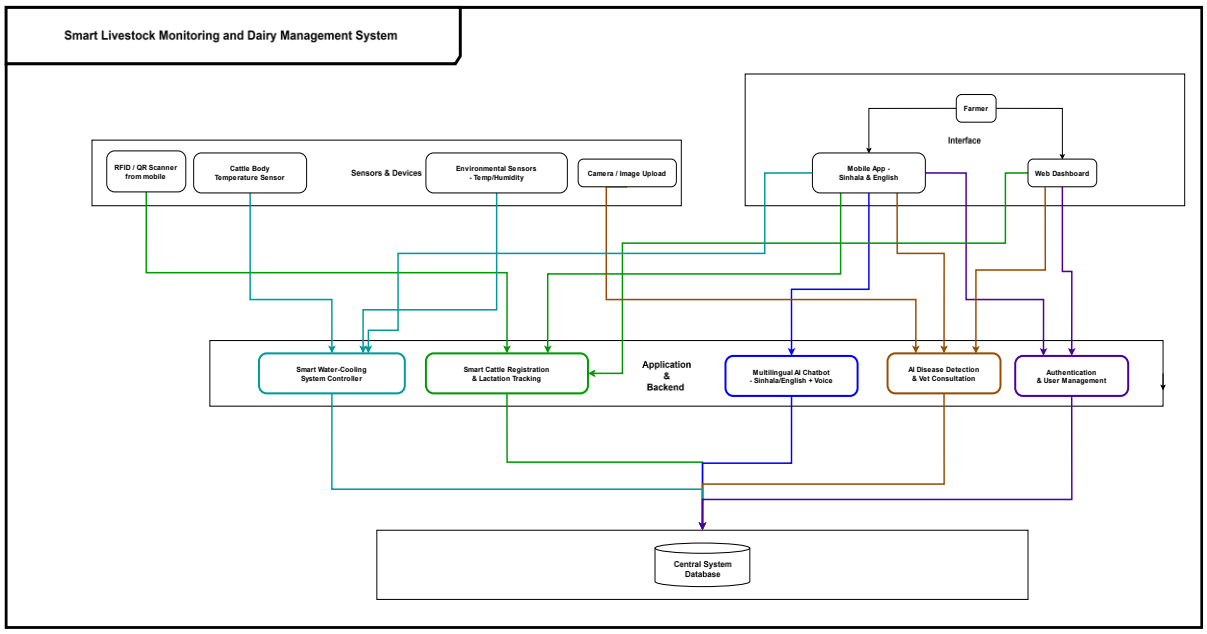
1. **Smart Real-Time Lactation Monitoring System** that tracks Multiple data to generate personalized lactation curves for each cow. It detects performance issues and gives real-time suggestions to farmers. This helps improve cow productivity, reduce extra costs, and support better decision-making for feeding and breeding making dairy farming more efficient and profitable.

2. The **Multilingual Farmer Education Chatbot** offers hands-free, voice-enabled guidance in Sinhala. It provides practical tips on animal care, disease symptoms, feeding practices, and hygiene management tailored to local conditions.

3. **Veterinary Assistance & Disease Detection with an AI-powered multimodal system** that identifies visible symptoms of diseases such as Foot-and-Mouth Disease and Lumpy Skin Disease using image recognition, text input, and lab report analysis. It also supports remote consultations with certified veterinarians through integrated chat and media sharing.

4. The **Smart Water-Cooling System with AI-Powered Wearable Neck Collars** uses for measure both body temperature and ambient temperature/humidity. These devices will send real-time data to a central gateway. Using AI/ML model we will predict individual heat stress levels. If a significant number of cows are in high stress, the system will automatically activate barn-level sprinklers using relay-controlled valves.

By combining these innovations, the platform helps farmers overcome traditional manual processes, reduce errors, and increase productivity. It empowers rural farmers with **localized, tech-driven tools** to make smarter decisions and improve dairy operations sustainably.



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

1. Domain Expertise

This project demands interdisciplinary expertise in livestock management, veterinary science, agricultural technology, and computer science. Veterinary knowledge is essential for disease symptom identification, vaccination schedules, and cattle health best practices. Experts in dairy farming, especially in tropical climates like Sri Lanka, are required to design region-specific solutions, such as feeding and cooling strategies. In addition, AI and IoT system architects are needed to integrate computer vision, sensor networks, and real-time analytics into a user-friendly mobile/web platform.

2. Knowledge

The system's success relies on practical understanding of:

Cattle behavior and physiology to inform image recognition and sensor-triggered actions.
Local farming practices and socio-economic constraints faced by medium-scale Sri Lankan dairy farmers.
Multilingual NLP to design a Sinhala-English chatbot accessible to low-literacy users.
Machine learning and computer vision techniques for disease detection.
IoT systems integration, including sensor calibration, data logging, and actuator control for cooling systems.

3. Data Requirements

To train and refine the system, key data sets include: Annotated images of cattle showing visible symptoms (for diseases such as FMD, LSD, Black Quarter).
Environmental and cattle temperature data for predictive cooling.
Historical health and lactation records of Sri Lankan cattle breeds.
Farmer interaction logs with chatbots to improve NLP accuracy.
Data must be localized, privacy-compliant, and representative of Sri Lanka's climatic and agricultural diversity.

8. Objectives and Novelty

Main Objective: Create a comprehensive Smart Dairy Farm Management Platform that seamlessly integrates livestock monitoring, cattle health management, farmer education, and automated environmental control systems. It is specifically designed to support **medium-scale farms in Sri Lanka**, where traditional, manual processes often limit productivity, health outcomes, and operational efficiency.

Member Name with Registration No	Sub Objective	Tasks	Novelty
Kamburugamuwa K.V.M.S. IT22885128	Develop individualized parametric or semi parametric lactaion curves for each cattles using models such as Wood's or Wilmink, based on parameters including Daily amount of milk an animal provide,Feed intake, Health Records and breeding status . The curves will be dynamically updated in real-time as new data is collected, enabling the system to detect	Collect daily data for each cow, including Daily amount of milk an animal provide (milk yield), feed intake, health records such as infections, and breeding details such as artificial insemination dates and calving history. This data will form the base for personalized lactation analysis. Build and Train model to generate a parametric or	Most existing systems that track milk production only at the herd level[1], this system focuses on individual cows, generating personalized lactation curves that capture each animal's unique milk production pattern. By integrating daily data on milk yield, feed intake, health, and breeding status, the system offers real-time, dynamic performance tracking rather than static historical summaries. The problem it aims to solve is the lack of real-time lactation monitoring and suggestion systems available to farmers. Currently, farmers have no effective way to identify important performance issues such as peak milk production times, unexpected drops, or prolonged dry periods. This limits their ability to make timely decisions that could enhance profitability. By providing early

	anomalies early and provide predictive alerts for potential performance issues such as delayed peak, persistent drop, or extended dry periods. Additionally, the system will deliver actionable insights to farmer, supporting decisions on Feed Optimization: Adjust rations if peak yield or persistency drops Breeding Decisions: Identify cows with superior curves for artificial inseminations Culling Suggestions: Detect consistent low performers for sale/meat	semi parametric lactation curve for each individual cow for show milk production changes over time. Build user interface to read and get the suggestions easily and understandlble way for farmers Train model to detect problems such as delayed peak production, a sudden drop in yield, or long dry periods. and provide suggestions based on the issue detected. Continuesly Tarin to give more smart suggestion	detection of these issues, the system helps farmers take necessary actions to optimize feeding, improve breeding decisions, save on unnecessary feed costs, and increase milk income. Ultimately, the system provides a practical, data-driven tool for individualized cow management, boosting farm efficiency and profitability while optimizing resource use. References – [1]M. Lee, S. Lee, J. Park, and S. Seo, "Clustering and Characterization of the Lactation Curves of Dairy Cows Using K-Medoids Clustering Algorithm," Animals, vol. 10, no. 8, p. 1348, Aug. 2020, doi: https://doi.org/10.3390/ani10081348.
Waduge S.U. IT22001184	Develop a Sinhala chatbot specified for Diary farm management with voice support to deliver localized guidance and tips.	Collect relevant domain-specific information related to dairy farming including feeding, milking, diseases, breeding, and seasonal care.	Implement a Sinhala chatbot with voice assistance specified for dairy farm. -Although multilingual chatbots with voice assistance have been successfully implemented in various areas, this project focuses on customizing such a system specifically for the dairy farming

		Translate and localize the collected content into Sinhala. Train a custom NLP model using platforms like Rasa or Dialogflow by feeding intents, training phrases, entities, and appropriate responses. Build a user interface that supports Sinhala Unicode text with an option for language switching. Create a chatbot backend that connects the NLP model to the knowledge base and retrieves answers based on matched intents. Integrate Sinhala-compatible speech-to-text (STT) functionality to accept voice queries from farmers and convert them into text input.	sector. The developed chatbot is tailored to address the unique needs of dairy farmers by providing localized guidance and tips in Sinhala, along with integrated voice support. This allows farmers to interact with the system using spoken input and receive voice responses, making it more accessible and practical, especially for users with limited literacy or digital skills. References- [1] Wap Avishka, Banujan Kuhaneswaran, and Hn Gunasinghe, "A Novel Conceptual Chatbot Architecture for the Sinhala Language – A Case Study on Food Ordering Scenario," Feb. 2022, doi: https://doi.org/10.1109/icarc54489.2022.9753725. https/ieeexplore.ieee.org/document/10374040 [2] Arjon Turnip, W. Nur, Nguyen Le Hoa, Abdi Dharma, None Ubaidillah, and Endra Joelianto, "Interactive Development of Medical Robot Using Voice Chatbot Based on Deep Learning," Nov. 2023, doi: https://doi.org/10.1109/ict60153.2023.10374040. [3] K. Herrera, J. Miranda, and D. Mauricio, "Milchbot: App to Support the Process of Feeding and Caring for Dairy Cows in Peru," *AGRIS on-line Papers in Economics and Informatics*, vol. 14, no. 4, pp. 27–

		Route the converted text through the chatbot logic to retrieve a response based on detected intent. Use a text-to-speech (TTS) engine that supports Sinhala to provide spoken responses back to the user. Continue maintaining and updating the knowledge base as new guidance and farming practices are introduced.	37, Dec. 2022, doi: https://doi.org/10.7160/aol.2022.140403.
Gunarathna P.D.A.G. IT22306586	Design a wearable IoT collar device for each cow that can measure both body temperature and ambient temperature/humidity. These devices will send real-time data to a central gateway. Using AI/ML model we will predict individual heat stress	Develop a wearable device (neck collar) for each cow using a DS18B20 sensor to collect body temperature and a BME680 sensor to gather environmental temperature and humidity data. Implement a gateway (using ESP32) that receives	This project brings a unique approach to cattle heat stress management by combining wearable IoT devices with AI-powered decision-making. Unlike existing systems that rely only on ambient temperature or spray water on a fixed schedule, our system monitors both the body temperature of individual cows and environmental conditions using smart collars. Machine learning is used to accurately predict heat stress levels in real time, allowing the system to control barn-level sprinklers only when needed. This ensures targeted cooling, reduces

	levels. If a significant number of cows are in high stress, the system will automatically activate barn-level sprinklers using relay-controlled valves. And also provide interface for real-time monitoring cattle data system status, alerts, and manual control options for farmers.	sensor data from multiple collars, processes it, and forwards it to the AI model. Build and train a machine learning model using features such as body temperature, ambient temperature, humidity, and time to classify cattle heat stress into levels (Low, Moderate, High). Use a relay module to automatically switch the water sprinklers on/off based on heat stress predictions. Sprinklers are triggered only when a significant percentage of cows in a barn show signs of stress. Implement an app (or web dashboard) for real-time monitoring. Farmers can view sensor data, receive heat stress alerts, check historical trends, and	unnecessary water use, and improves cattle comfort. The integration of a mobile/web dashboard also gives farmers real-time data and control, making the solution practical and user-friendly. References – [1] E. Liu, L. Liu, Z. Zhang, M. Qu and F. Xue, "An Automated Sprinkler Cooling System Effectively Alleviates Heat Stress in Dairy Cows," *Animals*, vol. 14, 2024, Art. no. 2586, doi: **https://doi.org/10.3390/ani14172586** [2] S. Goswami, "Implementation of Automatic Cooling System for Cattle Shed and Its Effect on Milk Production," *Indian Research Journal of Extension Education*, vol. 21, no. 2&3, pp. 137–143, Apr.–Jul. 2021 https://api.seea.org.in/uploads/pdf/2021-66-137-143.pdf

		control sprinkler systems manually if needed.	
GALLAGE D.S. IT22309310	Developing a System for Early Detection and Severity Assessment of Contagious Cattle Diseases, Including Foot and Mouth Disease (FMD) and Lumpy Skin Disease (LSD), Using a Multimodal AI Approach that Combines Image Inputs, Lab Report PDFs, and Text-Based Symptom Descriptions	1. Data Collection Farmer uses the mobile app to submit: An image of the affected cow (showing skin, mouth, legs, etc.) A lab report (as a PDF/image) A short text description of the symptoms (e.g., fever, limping, drooling) 2. Preprocessing -Image Processing: Uses a CNN model to detect visible signs. -Lab Report Analysis: Optical Character Recognition (OCR) is used to convert lab report images into text. Extracts values such as white blood cell (WBC) count, and PCR test results	The novelty of this proposed system lies in its multimodal AI approach for the early prediction of Foot and Mouth Disease (FMD) and Lumpy Skin Disease (LSD) in cattle, using a combination of three input types: images, lab report PDFs, and text-based symptom descriptions. Unlike most existing systems that rely on a single data type, this solution combines multiple modalities to achieve higher prediction accuracy. It also provides a severity assessment (mild, moderate, severe), which is rarely addressed in prior research. Importantly, both FMD and LSD are highly contagious diseases, and early identification can significantly reduce the risk of spreading infection across the herd. The system will be trained using datasets specific to local cattle, improving its effectiveness for local conditions and breeds. Additionally, veterinary integration ensures expert validation of AI predictions, making the system not only intelligent but also practical for real-world deployment in local dairy farming. References - [1] T. Velugoti, S. B. Dasari, and S. S. Sultana, “Lumpy Skin Disease Detection Using Deep Learning,” *2024 IEEE Students Conference on Engineering and*

		-Text Processing (Natural Language Processing NLP): Tokenizes the symptom text to Identify keywords 3. Multimodal AI Analysis: Combine all extracted features (Visual, Numerical and categorical lab results, Text-based symptom information). These are fed into a combined deep learning model, the model outputs a prediction based on the combined inputs. 4. Health Status Detection: The system first decides whether the cow is Healthy/ Unhealthy. 5. Disease Classification If unhealthy, check for: -Foot and Mouth Disease (FMD) / Lumpy Skin Disease (LSD) / If neither disease is detected, the case is flagged as "Other illness"	*Systems (SCES)*, pp. 1–6, Jun. 2024, doi: https://doi.org/10.1109/sces61914.2024.10652331. [2] Rajasri Maganti, S. Yalamanchili, and Balaji Mediboyina, "Cattle Skin Diseases Classification Using Deep Learning Models," pp. 698–703, Nov. 2024, doi: https://doi.org/10.1109/icdcc62744.2024.10961322. [3] R. Newlin Shebiah and S. Arivazhagan, "Deep Learning Based Image Analysis for Classification of Foot and Mouth Disease in Cattle," pp. 701–705, Aug. 2023, doi: https://doi.org/10.1109/icirca57980.2023.10220765.

		6. Severity Assessment: Labels the severity as Mild/ Moderate/ Severe 7. Output to Farmer: Disease name, Severity level, Explanation of findings (e.g, “nodules detected on neck and limbs with fever and positive PCR”), Advice for animal care (e.g, isolate animal, give fluids, clean wounds) 8. Veterinary Assistance: Each case is forwarded to a veterinarian to confirm or reject the AI prediction and use it as supporting input for the final diagnosis.	

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Kamburugamuwa K.V.M.S. IT22885128	This system focuses on monitoring individual cows by collecting daily data like milk yield, feed intake, health conditions, and breeding events. Based on this, it builds a personalized lactation curve to help farmers understand production changes over time and spot issues like yield drops or delayed peaks. Specialization Compliance: This component reflects core Software Architecture principles by using a modular setup with separate layers for data collection, curve generation, and suggestion output. It applies concepts from High-Performance Computing and Algorithm Design by efficiently processing and analyzing time-series data per cow. Mobile Computing and Rich Web Applications are integrated through a responsive interface, accessible via both web and mobile. The UI/UX design focuses on simplicity — showing clear graphs, alerts, and actionable suggestions, allowing farmers to interact with complex data effortlessly. Quality Assurance (QA) was ensured by unit testing the data input, curve generation logic, and suggestion outputs. The system uses structured error handling and maintains accurate historical data per cow, aligning with software engineering best practices around modularity, validation, and user-centered design
Waduge S.U. IT22001184	Multilingual custom knowledge-base chatbot with Sinhala support and voice assistance to educate farmers on cattle care, feeding practices, and vaccination schedules. Specialization Compliance: This component demonstrates the Software System through the implementation of a Natural Language Processing (NLP) model trained on localized phrases, mapped to intents using ML algorithms. Software Architecture is reflected in separating the chatbot engine, language modules, and voice input/output layers.

	The UI/UX design focused on inclusivity making the interface simple for rural farmers with limited literacy by adding voice-based input and responses in Sinhala. The system supports mobile and web integration, using REST APIs and frontend libraries to ensure cross-platform performance. QA concepts such as input sanitization, fallback responses, and intent testing were used to ensure chatbot accuracy and stability.
Gunarathna P.D.A.G. IT22306586	An IoT-based smart cooling system using AI-powered wearable collars that monitor individual cattle body temperature and environmental conditions to automatically control barn-level water sprinklers, effectively reducing heat stress and improving cattle welfare. Specialization Compliance: This component leverages Embedded Systems Architecture and IoT by integrating body temperature sensors (DS18B20) and environmental sensors (BME680) into ESP32-based wearable neck collars. These collars wirelessly transmit real-time data to a central gateway, which uses a trained machine learning model to analyze cattle-specific and ambient conditions. This demonstrates core skills in low-level system integration and wireless IoT communication. An intuitive mobile/web-based interface is developed using React Native and Firebase, applying UI/UX principles for real-time monitoring and control. The interface displays barn conditions, individual cow stats, stress alerts, and allows manual override of the sprinkler system—designed with clarity and responsiveness for ease of use in rural farm environments. The system incorporates AI-driven prediction algorithms (e.g., Random Forest or TinyML), replacing simple threshold logic with intelligent stress forecasting, aligning with Algorithm Design & Development (ADD) specialization. Quality Assurance (QA) includes calibration of sensor data, edge-case testing of stress predictions, and simulation under diverse weather conditions. The system supports cloud syncing and remote updates, reflecting modern software deployment and reliability engineering practices.

GALLAGE D.S. IT22309310	This AI-based multimodal disease detection platform identifies Foot-and-Mouth Disease (FMD) and Lumpy Skin Disease (LSD) in dairy cattle by analyzing images, farmer-reported text symptoms, and PDF lab reports using computer vision, NLP, and OCR techniques. The system architecture is modular, separating model training, prediction, and veterinary consultation components. A mobile interface enables easy image upload, symptom entry, and lab report submission, designed with a user-friendly UI for rural farmers. Disease detection combines predictions from all input types, improving accuracy and early diagnosis. Quality assurance includes data validation, error handling, and model evaluation using accuracy, precision, and recall metrics.

10.Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Ms.	Karthiga	Rajendran	KR 2025/06/26 (for Ms. Karthiga)
Co-Supervisor	Ms.	Kaushalya	Rajapakse	KR 2025/06/26
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

IT4010 Research Project – 2025 July

Topic Assessment Form

<u>This part is to be filled by the Topic Screening Staff members.</u>

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	

b) Does the proposed topic exhibit novelty?

Yes		No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

IT4010 Research Project – 2025 July

Topic Assessment Form

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

*__Important__:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.

2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.